DECODELABS INTERNSHIP — PROJECT 1

Data Cleaning & Preparation

PROJECT OVERVIEW

This project was completed as Project 1 of the Data Analytics Internship at DecodeLabs. The main objective was to clean, transform, and prepare raw data for accurate analysis by identifying missing values, removing duplicate records, standardizing data formats, and performing rigorous data quality checks.

TOOLS & TECHNOLOGIES

`Python` `Pandas` `NumPy` `Google Colab` `OpenPyXL` `ReportLab`

PROJECT TASKS

1. **Data Inspection:** Examining dataset structure, columns, data types, and statistical summaries.
2. **Missing Values:** Identifying missing values and applying appropriate imputation strategies, including mean, median, and mode.
3. **Duplicate Removal:** Detecting and removing duplicate records and duplicate unique IDs.
4. **Date Standardization:** Converting date values into ISO format (`YYYY-MM-DD`).
5. **Text Standardization:** Removing extra spaces and standardizing text capitalization.
6. **City Standardization:** Cleaning inconsistent city names.
7. **Numeric Precision:** Rounding numeric values to two decimal places.
8. **Data Validation:** Performing final integrity and quality checks.
9. **Change Log:** Generating a PDF report documenting data cleaning operations.

PROJECT STRUCTURE

```
DecodeLabs-Data-Cleaning-Preparation/
│
├── README.md
├── data/
│   └── raw_dataset.csv
├── notebooks/
│   └── Data_Cleaning_Preparation.ipynb
├── output/
│   ├── cleaned_dataset.csv
│   ├── data_cleaning_changelog.pdf
│   └── audit_report.csv
├── requirements.txt
└── .gitignore
```

INSTALLATION & EXECUTION

Installation: Install the required Python packages:

```
pip install -r requirements.txt
```

How to Run:

1. Clone or download this repository.
2. Open the notebook in Google Colab or Jupyter Notebook.
3. Upload the raw dataset when prompted.
4. Run the notebook cells sequentially.
5. Review the cleaned dataset, PDF change log, and audit report in the `output/` directory.

OUTCOMES & KEY LEARNINGS

Expected Outcomes: Improved data consistency and accuracy, proper handling of missing values/duplicates, standardized fields, and a final data quality audit.

Practical Experience Gained: Data preprocessing and cleaning, core Pandas & NumPy manipulation, field standardization, quality validation, and automated PDF report generation.

INTERNSHIP METADATA

Organization	DecodeLabs
Program	Data Analytics Internship
Project	Project 1 – Data Cleaning & Preparation

> *This repository is intended for educational and portfolio purposes. The practice dataset is synthetic and does not represent real customer transactions.*